

Final report for grant A42, design of the Big Quiet Fan.

A fan to greatly outperform the existing fan solutions for air purifiers.

The project went remarkably well and most of the progress was made in a very short time. Later stages did take longer than I would like, mostly due to waiting on others which tends to be extremely time consuming in the world of manufacturing for some reason.

Extremely good performance has been achieved. Energy efficiency, noise to flow/pressure relationship and even cost of manufacture are all very very good now. Progress was repeatedly made in a less than expected way. Several times I thought I was basically done but was able to make progress on core metrics, notably the CADR to noise ratio when fitted with a typical filter set (about 1000 CFM CADR at 9 pascals of pressure, usually 4 1x25x20 MERV-13 Filtrete filters plus a 20 by 20 by 1 inch one on the bottom of the box, with legs).

I discovered several basic techniques to improve performance of the blades, which can be applied to many other fans in the future:

- Increase the depth of the blades to increase pre-swirl and increase pressure/flow generation of the first blades (what I call the primary blades).
- Use a similar stator (the secondary, I call it), which uses the pre-swirl to boost flow/pressure. This is usually called a straightener, and it does straighten the airflow, but it's purpose is not actually to do so. I realized it's basically a fan blade, except the air rotates relative to the blades while the blades stay still.
- Again the stator should be relatively deep
- The curvature of the blades needs to be optimized for any given fan, which I have done here. The optimal in this case appears to be a sort of spiral in the plane perpendicular to the axis of rotation, and a parabolic curve in the other plane.
- The number of blades for both primary and secondary also needs to be optimized, this of course dictates spacing.
- The spirals of the primary and secondary should be of opposite handedness, to eliminate buzzing noise. Buzzing noises can still arise depending on the details of the geometry, I was able to find some combination that practically eliminates any buzz.
- As usual the root of the blades has a different average angle of attack than the tip, usually higher.
- An outermost cylindrical wall allows the blades to be thin and light, and any geometry desired as their stiffness is not a problem. It may help with the turbulence at the tips but

I don't actually know because it is probably at least harmless and useful to improve the mechanical properties of the blades, so I never tried without it.

- Use a high damping plastic/material for the secondary to reduce noise, this helps a surprising amount.

The process of optimizing the curves and number of blades etc. is basically the hard part. I used simulations but due to the large size of the object and the detail required that didn't give very good accuracy. The limits of the software I was using were too severe, but certainly it's fundamentally possible to get better results. In general the software can tell the turbulence energy which should correlate strongly with noise.

One thing that has potential but I did not try much is to change the curvature of the front and back of the blades. In general, I use a thin sheet of material, the curve being the same on both sides. In general an airfoil works better with a different curve on the back than the front, however this would be very heavy and consume more plastic when molded, it also takes far longer to print. I did experiment a bit and it seemed unpromising.

It is probably possible to slightly increase performance by maybe 10 percent, but progress really slowed down and that would be a bit costly. I pretty much hit the budgeted \$ on the head so I feel I struck a good compromise.

In general, I measured the pressure inside the filter box while testing a wide range of primaries and secondaries, and each promising combination of the two. I did more than 65 tests of unique combinations. I also poked around visualizing the flow with fog, and hypothesized how promising ways to improve things, combining the best features.

One feature that was promising in simulation but did not help in reality was the use of vertical cylindrical walls that divide up the blades, allowing the blade spacing to be optimized in the different regions. The thing is the blades are moving through the air more slowly in the middle than the edge so it would stand to reason the optimal geometry and spacing would be different there. It also interrupts the undesired radial flow of the air due to centrifugal force. Although this type of feature did improve the rpm to pressure/flow relationship, they increased noise unexpectedly so were dispensed with.

I realized several things:

- Most of the buzzing noise is from the pressure wave hitting the secondary. The opposite handed spirals is the best way to solve this as it also works well in other ways. However optimization of the curves is needed to really polish it off.
- The spacing between the primary and secondary is also very important. In this fan it should be between 3-4 mm. A bit outside that range isn't a big deal, but it does adversely affect noise/flow/pressure relationship.
- Bearing noise is actually significant and Z4 bearings should be used.

I also found a clever solution for the motor drive system which has many other applications where extremely smooth and quiet motion is desired, especially with a high torque to speed ratio at a low cost:

- The stepper motors were fundamentally promising as they have good torque/speed ratio and are low cost and easy to get. I was actually able to source a hall effect sensor equipped gimbal motor which they say could be supplied for \$45 USD, which is highly unusual for such motors of such size and properties, a very good deal. However that came too late and I never even tested it. A suitable stepper is about \$8 and the supply is inherently more reliable, so it's better to use that if you can.
- I knew that sine wave drive tends to work pretty well, audible ring and other noises typically involved with stepper motors can be eliminated, the only remaining vibration would be cogging. The other challenge was commutation while keeping cost low.
- I was able to re-invent synchronous drive basically, almost by accident. It turns out if you just ramp the voltage with RPM that actually works great to drive the motor, the current and torque is automatically regulated within a fairly wide range due to the way the back emf changes with the so called torque angle (when the motor is loaded more, back emf goes down allowing more current, increasing torque to match the load).
- However I found the system oscillated too much, leading to step loss. I was able to solve this with damping using TPU, a flywheel made of TPU with enough bolts around it's periphery for mass was sufficient.
- The last bit of vibration was dealt with by isolation, using a flexible coupler. It was difficult to deal with the creep set etc. however I was able to find a solution.
- The noise from even the Z4 bearings (quietest you can get) had to be isolated with a small hard TPU cushion (72D) around the bearings, but this is cheap and works well.

Unfortunately this whole system does imply increased costs over a bare stepper motor directly driving the blades, but not much, just some plastic and rubber parts and a couple bearings, and the shaft. The parts can all be printed or molded easily.

I did not try multiple rotors, perhaps on the same shaft. That has potential, it increases size but could be sensible. That would be the next major pattern to employ to further increase the noise/cadr ratio, I think. However it increases size and complexity and thus cost. This system is quite close to any ordinary fan, in its simplicity, and I think it's rare you would have a compelling need for better CADR/noise ratio. You can always just use twice as many air purifiers, in a pinch. I think this is a good compromise.

Driving the fan blades with the motor directly does actually work, the vibrations are damped enough, but you get a buzzing noise. I tried quite extensively to eliminate this buzzing noise through firmware optimization. Theoretically I thought the cogging could be eliminated by superimposing the waveform 180 degrees out of phase, causing a torque ripple that cancels out the cogging or other torque ripple. I built a system that was able to measure torque ripple normatively and used a global optimizer to adjust the waveform, searching for any waveform that would help and improving on it. The system was good, if I imposed a torque ripple it was able to quickly find the right waveform to cancel it out, however for some reason that I still do

not understand it was never able to make meaningful reductions in the torque ripple that the stepper motor exhibited with pure sine wave drive. It may be that there were oscillations in the electrical-mechanical system combined which were dynamic in origin. However I did actually test for this by putting damping in in the electrical line and it didn't seem like it. It's possible a damping flywheel on the back of a dual shaft stepper with the blades directly mounted to the motor would work, but I don't think so. I tried a rubber coupler on the blades which should have a similar effect and it hardly helped.

I have designed a drive system PCB which is applicable to other applications and can easily be reprogrammed. It uses relatively common and low cost parts. It is fairly refined, I got the pro layout service and have gone through several prototypes. Extra pins are broken out for auxiliary use etc. The Arduino program is also of course published and can be adapted to other scenarios.

Budget:

Total budget was \$50,000 USD for the design and development, including all materials, equipment, premises and labor. Most of this was expended and the remainder has been woven into future grants.

					cad	usd
labor rate					35	
					CAD	USD
labor cost					44800	32463.76812
Parts/equipment /materials					15831	11471.73913
premises					1800	1304.347826
With labor					62431	45239.85507

Estimated impact

This part of the road has now been well paved for far superior noise/cadr ratio air purifiers for millions or even billions of people around the globe. To have not undertaken such a research project would have been absurd, given the number of fans and air purifiers that are needed across the globe in the coming years.

I would argue it is a bit wacky to only have one person on a relatively tight budget doing this, given the apparent potential. It would be great to build a simulation system and really make something completely optimal. However I would be surprised if that led to even a 10% improvement in CADR to noise ratio.

This fan as it stands really is pretty good. I can turn it up to 400 rpm (1030 cfm CADR measured) and work in my apartment right next to it without irritation, in fact the sheer amount it moves around, causing quite a draft is the most annoying part of leaving it on like that. If I turn it down it is inaudible at normal distances even with 36 dBa background noise, allowing tremendous CADR while sleeping.

The techniques could also be applied to other smaller format air purifiers. There are companies that make 250 mm PC fans for instance, it would be a good idea to use these tricks in that context, especially combined with their low cost approach to the motor.